EE P 564: TinyML
Lab 9: Responsible AI in TinyML

Dept. of Electrical and Computer Engineering
University of Washington

Instructor: Dinuka Sahabandu (sdinuka@uw.edu)

ELECTRICAL & COMPUTER ENGINEERING
UNIVERSITY of WASHINGTON

NSL

Today’s Lab

- Both Software and Hardware Lab!
- Training models using TensorFlow-Privacy
- Converting model to TensorFlow Lite format
- Deploy the model to the Arduino board

Lab 9 – Software - Road Map

1. Data Processing

2. Model training using TensorFlow-Privacy

3. Model Evaluation

4. Covert the TensorFlow model to

TensorFlow Lite Micro model

5.Compress the model

Lab 9 - Hardware - Road Map

1. Connect Arduino Boards to your computer

2. Implement converted ML model

3. Deploy the model on to board

4. Open Serial Monitor and test the model

Training the Model with Differential Privacy

Open EEP564-TinyML-Lab9.ipynb in Google Colab to train the model with differential privacy and convert it into a hex array format for deployment.

🧠 Train your gesture classifier with Differential Privacy

- Preprocess and normalize the same gesture dataset from Lab 8
- Use `tensorflow_privacy.DPKerasSGDOptimizer` to train a model with privacy-preserving updates
- Observe training and validation performance over 500 epochs
- Export the model as `gesture_model.tflite`

🔄 Convert the model to TensorFlow Lite and generate header file

- Convert the model to `.tflite` format using `TFLiteConverter` (no quantization)
- Use `xxd` to convert `gesture_model.tflite` into a C header file named `model.h`
- This header file contains a hex array version of the model

Included in EEP564-TinyML-Lab9.ipynb

Deploying the Model with Differential Privacy

📥 Replace model in Arduino deployment

- Open your Lab 8 Arduino `.ino` file in Arduino IDE
- Replace the hex array in Lab 8's `model.h` with the new array from Lab 9's `model.h`
- Upload the sketch to your Arduino Nano BLE Sense

🖥️ Test the DP model on-device

- Open Serial Monitor
- Perform gesture inputs and observe predicted outputs
- Take a **screenshot of Serial Monitor** showing model predictions

Lab 9 Canvas Submission Details

📊 Compare results with Lab 8 (submit as PDF)

For your Canvas submission:

- Compare Lab 8 and Lab 9:
  - **Training performance** (loss, accuracy trends)
  - **Test accuracy**
  - **On-device predictions and latency**
- Discuss **how DP affects performance and generalization**
- Intuitively explain how DP helps prevent attacks that extract training data (e.g., membership inference)

📤 Submit as **one PDF** on **Canvas → Assignments → Lab 9**

Include:

- Screenshot of Serial Monitor
- Comparison + short write-up on DP effects